Overhead Line Insulators

Class Notes

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DAY # 7 [15/9/25]

Types of Insulators :

(1) Pin - used up to 33 kV

(2) Suspension / String - used for all Vtg - 33 to 765 kV

(3) Strain - used in road, railway, river crossings and dead end towers to reduce SAG.

(4) Post - used in S/S to support CT, PT, CVT

Overhead Line Insulators :

* Strength depends on : switching over vtg.

* Materials : Toughened Glass, Porcelain, Polymer

Suspension Insulators :

No. of insulator discs (n) = $\frac{\text{operating phase vtg of T/L}}{\text{Maximum withstand vtg of each disc}} = \frac{V_{ph}}{V_D}$

Assume V_D = 11 kV

V_L (kV) / n (no. of discs)

33 : $\frac{33/\sqrt{3}}{11}$ = 2 (3 to 5 discs)

132 : $\frac{132/\sqrt{3}}{11}$ = 7 (8 to 10 discs)

220 : $\frac{220/\sqrt{3}}{11}$ = 12 (13 to 15 discs)

String efficiency (η)

$$\%\eta = \frac{\text{Operating phase vtg of T/L}}{n \times (\text{Vtg of disc nearest to conductor})} \times 100$$

For n discs :

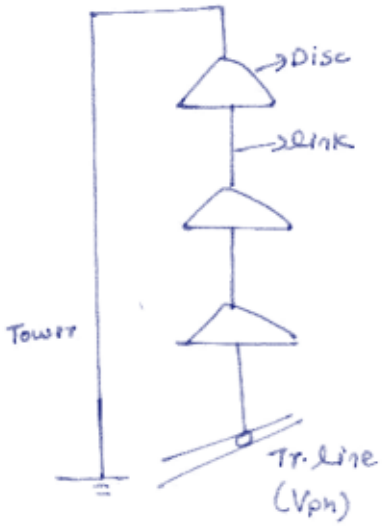
$$\%\eta = \frac{V_{ph}}{n \times V_n} \times 100 = \frac{V_1 + V_2 + V_3 + \cdots + V_n}{n \times V_n} \times 100$$

For 3 discs : if $V_1 = V_2 = V_3$, η = 100 %

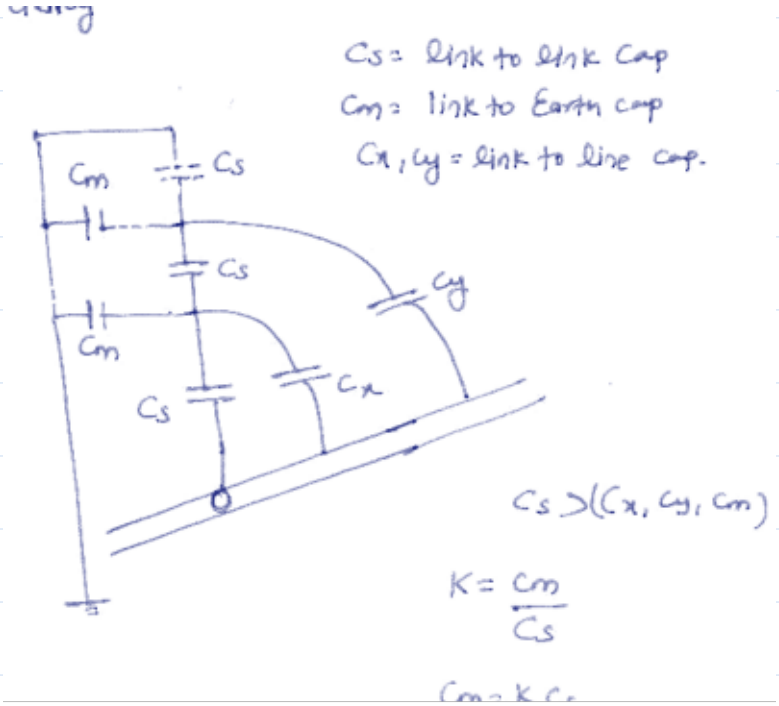
$$\%\eta = \frac{V_1 + V_2 + V_3}{3 \times V_3} \times 100$$

> Arcing horn : used to protect insulator disc from the back flash over vtg due to lightning surges.

Calculation of string efficiency



Tower, three discs joined by links, transmission line (Vph)

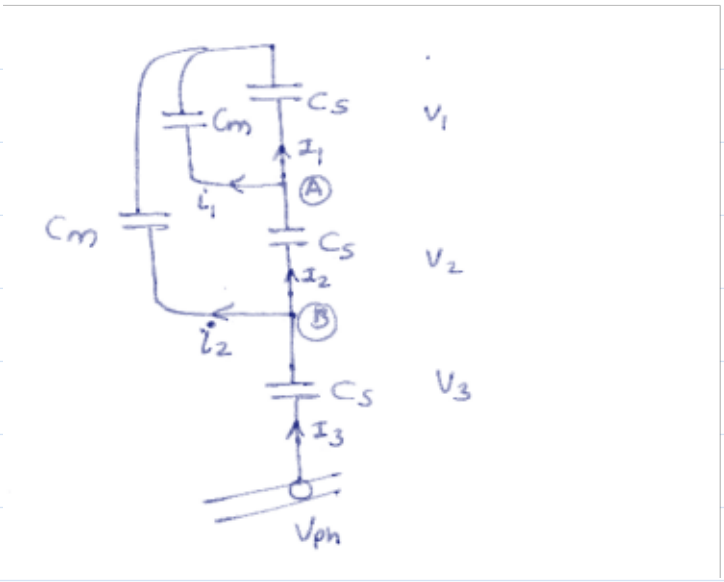


Equivalent: Cs = link to link capacitance, Cm = link to earth capacitance, Cx, Cy = link to line capacitance; Cs > (Cx, Cy, Cm)

C_s = link to link Cap ; C_m = link to Earth cap ; C_x, C_y = link to line cap.

$K = \frac{C_m}{C_s}$, $C_m = KC_s$

Case 1 : without C_x and C_y



Three discs with Cs, link-to-earth capacitances Cm at nodes A and B; currents I1, I2, I3, i1, i2; voltages V1, V2, V3; line Vph

At node A : $I_2 = I_1 + i_1$

$V_2 \omega C_s = V_1 \omega C_s + V_1 \omega C_m$; substitute $C_m = KC_s$

$$V_2 = V_1(1+K) \quad -(2)$$

At node B : $I_3 = I_2 + i_2$

$V_3\omega C_s = V_2\omega C_s + (V_1 + V_2)\omega C_m$ $-(3)$; substitute $C_m = KC_s$ and (2) in (3)

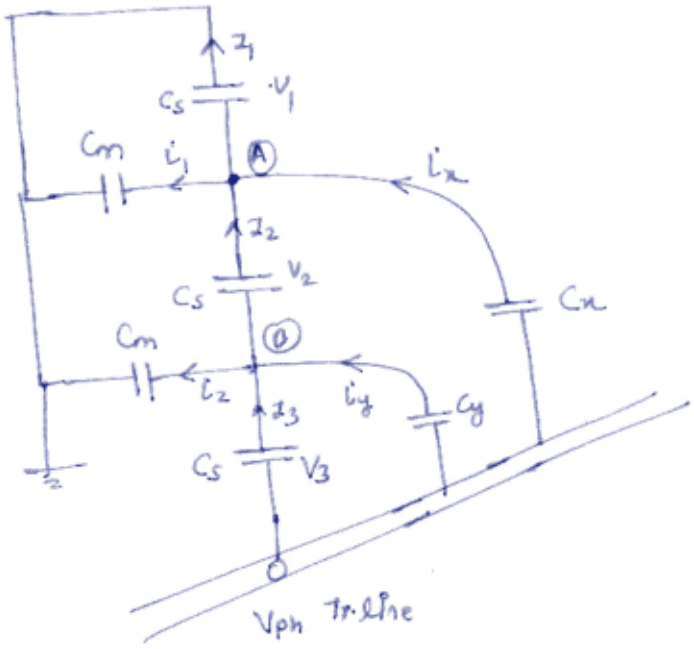
$$V_3 = (K^2 + 3K + 1)V_1 \quad -(4)$$

$$V_{ph} = V_1 + V_2 + V_3 = V_1 + V_1(1+K) + (K^2+3K+1)V_1 = (K^2+4K+3)V_1$$

$$\%\eta = \frac{V_{ph}}{3V_3} \times 100 = \frac{K^2+4K+3}{3(K^2+3K+1)} \times 100$$

If $K = 0$, $\eta = 100\ \%$

Case 2 : with C_x and C_y



Three discs with Cs, Cm to earth at nodes A and B, Cx and Cy link-to-line capacitances; currents I1, I2, I3, i1, i2, ix, iy; voltages V1, V2, V3; line Vph

Node A : $I_2 + i_x = I_1 + i_1$

$V_2\omega C_s + (V_2 + V_3)\omega C_x = V_1\omega C_s + V_1\omega C_m$

$$V_2(C_s + C_x) + V_3 C_x = V_1(C_s + C_m)$$

$$\frac{V_1}{\sqrt{3}}(C_s + C_m) - \frac{V_2}{\sqrt{3}}(C_s + C_x) = C_x \quad -(1)$$

Node B : $I_3 + i_y = I_2 + i_2$

$$V_3 \omega C_s + V_3 \omega C_y = V_2 \omega C_s + (V_1 + V_2)\omega C_m$$

$$V_3(C_s + C_y) = V_1 C_m + V_2(C_s + C_m)$$

$$\left(\frac{V_1}{\sqrt{3}}\right) C_m + \left(\frac{V_2}{\sqrt{3}}\right)(C_s + C_m) = (C_s + C_y) \quad -(2)$$

By solving (1) and (2) -> $\frac{V_1}{\sqrt{3}}$, $\frac{V_2}{\sqrt{3}}$

$$\%\eta = \frac{V_{ph}}{3\sqrt{3}} \times 100 = \frac{V_1 + V_2 + V_3}{3\sqrt{3}} \times 100 \text{ (with } V_3 = \sqrt{3} \text{ as reference)}$$

$$\boxed{\%\eta = \frac{1}{3}\left[\left(\frac{V_1}{\sqrt{3}}\right) + \left(\frac{V_2}{\sqrt{3}}\right) + 1\right] \times 100}$$

Eg: Calculate η of 3 discs, if $C_s = C$, $C_m = 0.2C$, $C_x = C_y = 0.1C$?

$$K = \frac{C_m}{C_s} = \frac{0.2}{C} \cdot C = 0.2$$

$$0.1C = \frac{V_1}{\sqrt{3}}[C + 0.2C] - \frac{V_2}{\sqrt{3}}[C + 0.1C] \quad -(1)$$

$$0.1 = \frac{V_1}{3}(1.2) - \frac{V_2}{3}(1.1)$$

$$1.1C = \frac{V_1}{\sqrt{3}}(0.2C) + \frac{V_2}{\sqrt{3}}(1.2C) \quad -(2)$$

$$1.1 = 0.2\frac{V_1}{3} + \frac{V_2}{3}(1.2)$$

$$V_1 = 2.385 \; ; \; V_2 = -0.3975$$

$$\%\eta = \frac{1}{3}[2.385 \ldots$$ (working left incomplete on the original page)

Eg: A 3-ϕ T/L is supported by 3 insulator discs in each phase. Vtg of the disc nearest to the conductor is 10 kV. Find the

(i) Vtg across each unit of the disc

(ii) Line vtg of T/L

(iii) If $Z_s = 400\,\Omega$ find SIL. Assume (K = 0.2)

(iv) If K = 0 find string η and vtg of each, SIL and operating line vtg with $Z_s = 400\,\Omega$.

(i) $K = 0.2$; $V_3 = 10$ kV

$V_3 = (K^2 + 3K + 1)V_1 \rightarrow V_1 = \dfrac{10k}{K^2 + 3K + 1} = 6.1$ kV

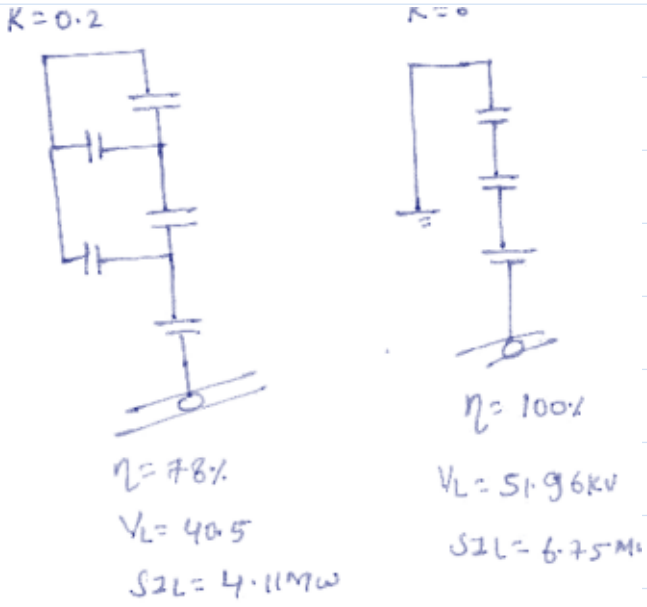
$V_2 = (1 + K)V_1 = 6.1(1.2) = 7.32$ kV

$V_{ph} = V_1 + V_2 + V_3 = 23.414$ kV ; $V_L = \sqrt{3}(23.414) = 40.5$ kV

$SIL = \dfrac{V_L^2}{Z_s} = \dfrac{(40.5)^2}{400} = 4.11$ MW

$\eta = \dfrac{K^2 + 4K + 3}{3(K^2 + 3K + 1)} \times 100 = 78\,\%$

(ii) $K = 0$; $\eta = 100\,\%$; $V_1 = V_2 = V_3 = 10$ kV ; $V_{ph} = 30$ kV ; $V_L = 30\sqrt{3} = 51.96$ kV ; SIL = 6.75 MW



K = 0.2: eta = 78%, VL = 40.5, SIL = 4.11 MW; K = 0: eta = 100%, VL = 51.96 kV, SIL = 6.75 MW

Advantage of Higher string efficiency :

(1) Uniform vtg can be maintained across all the discs

(2) Transmission vtg capacity can be improved

(3) Power T/F capability (SIL) can be improved

Methods of Improving string η :

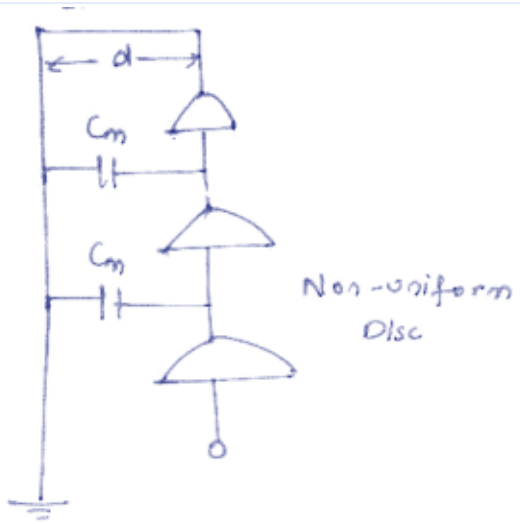
(1) By increasing the cross arm length the value of K can be reduced up to 0.1 so that η is improved (not practical)

(2) By using Non-uniform disc size (not practical)

(3) By using Guard ring or wire (most practical)

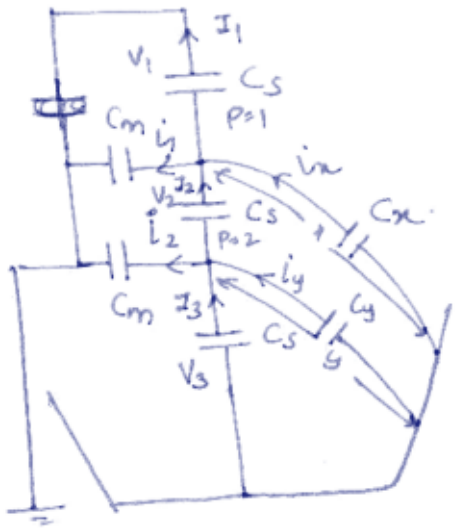
$$K = \frac{C_m}{C_s} = \frac{A\epsilon}{dC_s} ; \quad \epsilon = 8.85 \times 10^{-12}$$

$K = 0 \Rightarrow C_s = \infty$, $d = \infty$ (Not practical)



Cross arm of length d, Cm at each disc, non-uniform disc size

Guard ring or Guard wire :



String with guard ring: Cs, Cm, Cx, Cy, currents I1, I2, I3, ix, iy, i1, i2 and voltages V1, V2, V3; link numbers p = 1, 2

Node A : $I_2 + i_x = I_1 + i_1$; if $i_x = i_1$ then $I_2 = I_1$; $V_2 \omega C_s = V_1 \omega C_s$

$$V_2 = V_1$$

Node B : $I_3 + i_y = I_2 + i_2$; $i_y = i_2$; $I_3 = I_2$; $V_3 \omega C_s = V_2 \omega C_s$; $V_3 = V_2 = V_1$

$$\eta = 100\%$$

For η = 100 % : Capacitance of pth link

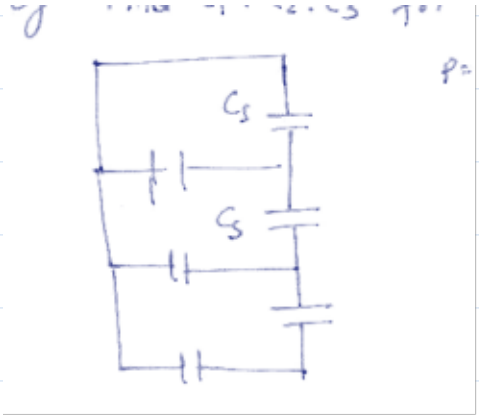
$$C_p = \frac{p C_m}{(n-p)}$$

p = link number ; n = No. of discs

At p = 1, n = 3 : $C_p = C_x = \frac{1 \cdot C_m}{3-1} = \frac{C_m}{2}$

At p = 2 : $C_p = C_y = \frac{2C_m}{3-2} = 2C_m$

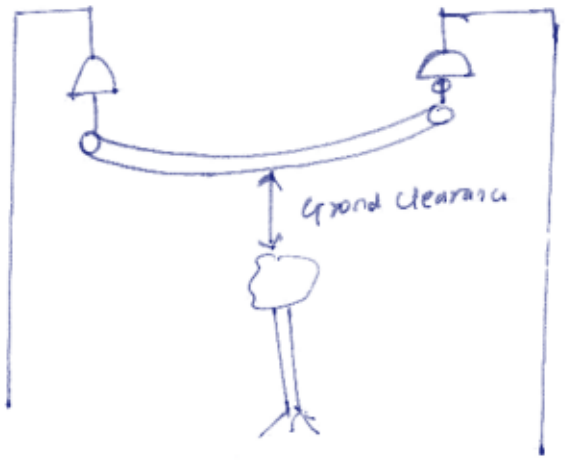
Eg: Find $C_1 : C_2 : C_3$ for η 100 % (p = 1, p = 2, p = 3)



Three-disc string with Cs and Cm per link, guard capacitances C1, C2, C3 at p = 1, 2, 3

Sag and Tension :

* Sag is the vertical distance from the point of support of the conductor to the lowest point.
* Sag is mainly produced due to (i) self wt. of ACSR conductor (ii) ice loading surrounding the conductor (iii) wind pressure on the conductor (iv) heating developed inside the conductor due to power and atmospheric pressure / temp change.
* Optimum Gnd clearance has to be maintained to avoid electrical S.C or electromagnetic induction due to sag in the line.



$S = \frac{w l^2}{8T}$

Conductor between two towers on insulator strings showing sag and ground clearance above a person/tree

$$S = \frac{w l^2}{8T}$$

V_L (kV) / Minimum Gnd clearance (m)

66 / 5.5

132 / 6.1

220 / 7

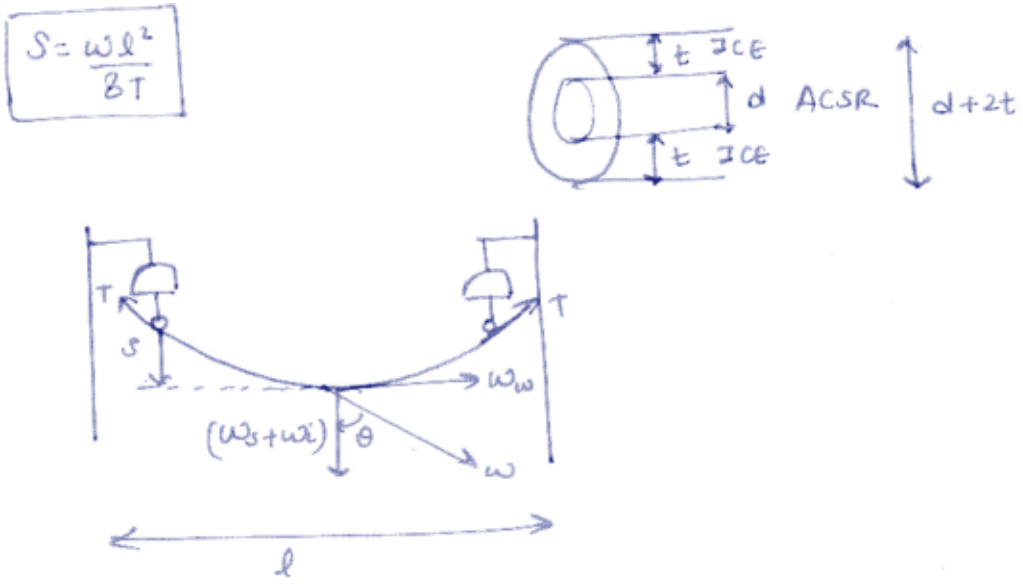
400 / 8.8

765 / 12.1

1200 / 15.4

(1) Towers are at same level :

$$S = \frac{wl^2}{8T}$$



Ice-coated ACSR cross-section (diameter d, ice thickness t, outer d + 2t); conductor sag S between towers at same level, tension T, loads (ws + wi) vertical, ww horizontal, resultant w at angle theta, span l

w_s = Self weight of ACSR per unit length

w_i = weight of ice loading per unit length = Density of ice x Volume of ice = Density of ice $\times \frac{\pi}{4}((d+2t)^2 - d^2) \times 1$ = Density of ice $\times \pi t(d+t) \times 1$

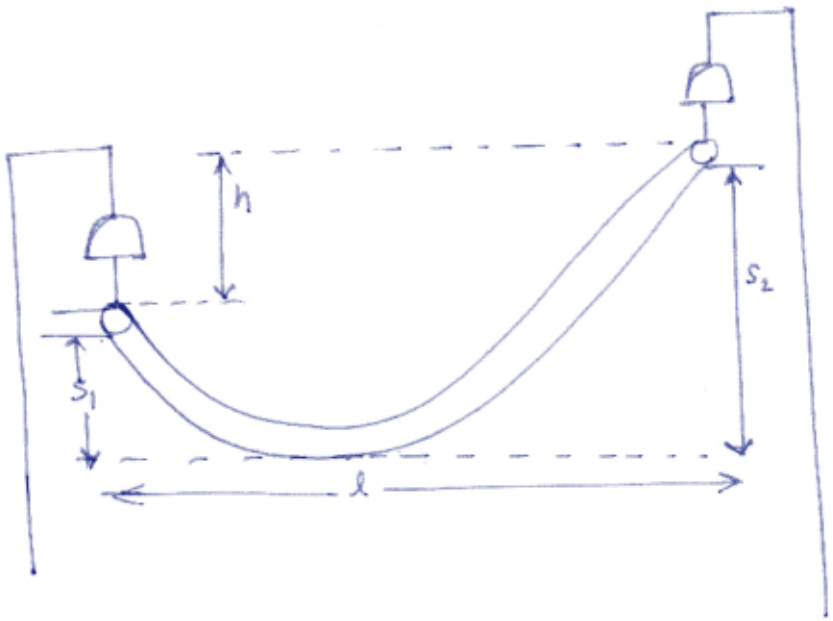
w_w = weight of wind per unit length = wind pressure per unit area x projected area per unit length = wind pressure per unit area x $(d + 2t) \times 1$ m

d = diameter of ACSR ; t = Thickness of ice.

$$Sag = S = \frac{wl^2}{8T} ; \qquad w = \sqrt{w_w^2 + (w_i + w_s)^2}$$

l = span length in m ; T = Tension = $\frac{\text{Ultimate strength}}{\text{Safety factor}}$

(2) Towers are at different height level : [Hill Area]



Two towers at different heights, height difference h, sags S1 (lower) and S2 (upper) measured from lowest point of conductor, span l

$$S_1 = \frac{w}{2T}\left(\frac{l}{2} - \frac{Th}{wl}\right)^2$$

$$S_2 = \frac{w}{2T}\left(\frac{l}{2} + \frac{Th}{wl}\right)^2$$

$$h = (S_2 - S_1)$$